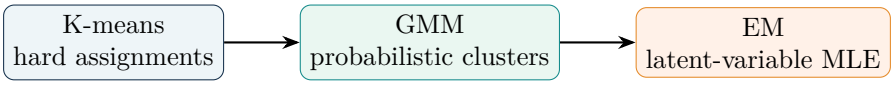
Clustering, Gaussian Mixture Models & the EM Algorithm

Refined Mathematical Lecture Notes

Study Notes



Abstract

These notes develop clustering from first principles. We first derive K-means as an alternating optimization method for squared-error distortion, then identify its modelling limitations and motivate Gaussian Mixture Models (GMMs). We derive the GMM likelihood and posterior responsibilities, and then develop Expectation-Maximization (EM) both as a practical algorithm for GMMs and as a general latent-variable maximum-likelihood method. The notes include geometric intuition, worked calculations, convergence caveats, the relationship between K-means and GMMs, applications, misconceptions, and a compact revision sheet.

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1 Clustering: the problem before the algorithm

Clustering is an **unsupervised learning** task: we observe feature vectors but do not receive class labels. Given

$$X = \{x_1, \dots, x_N\}, \qquad x_n \in \mathbb{R}^d,$$

we want to organize observations into groups whose members are similar under a chosen notion of distance or probability.

Key idea

A cluster is not automatically a "true class." The result depends on the representation of the data, the similarity measure, the model family, and its assumptions.

1.1 Where clustering is useful

- **Customer segmentation:** discover groups with similar purchasing patterns.
- **Image segmentation:** group pixels by colour, texture, or learned features.
- **Document analysis:** group documents by topic-related representations.
- **Biology and medicine:** explore subgroups of patients or measurements.
- **Compression:** represent many observations using a small set of prototypes.

1.2 Notation

We use N for the number of observations, d for feature dimension, and K for the number of clusters/-components. The symbol μ_k denotes a centre or Gaussian mean. In K-means, $c_n \in \{1, \dots, K\}$ is a hard cluster label; in a GMM, z_n is a latent component identity and r_{nk} is the posterior probability that point n came from component k.

2 K-means clustering

2.1 Core idea and geometric intuition

K-means represents each cluster by a single centroid. Every observation is assigned to its nearest centroid, and each centroid is replaced by the mean of the points currently assigned to it. These two operations repeat until assignments stop changing or improvement becomes negligible.

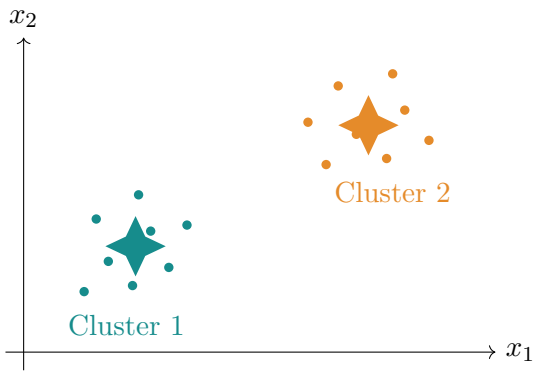


Figure 1: Illustrative clusters represented by centroids. The centroid is a mean, not necessarily an observed data point.

2.2 Objective function: within-cluster distortion

K-means seeks centres $\mu_1, \ldots, \mu_K$ and assignments $c_1, \ldots, c_N$ that minimize the total squared Euclidean distance:

$$\boxed{J(c, \mu) = \sum_{n=1}^{N} \|x_n - \mu_{c_n}\|_2^2.}$$

Equivalently, introduce an indicator r_{nk}:

$$r_{nk} = \begin{cases} 1, & c_n = k, \\ 0, & c_n \neq k, \end{cases} \qquad \sum_{k=1}^{K} r_{nk} = 1.$$

Then

$$\boxed{J(R, \mu) = \sum_{n=1}^{N} \sum_{k=1}^{K} r_{nk} \|x_n - \mu_k\|^2.}$$

The objective is a sum of squared distances from each point to the centre of its assigned cluster.

2.3 Deriving the assignment step

Suppose the centroids are fixed. For a particular point x_n, its contribution to J is minimized by choosing the centre with the smallest squared distance:

$$\boxed{c_n \leftarrow \operatorname*{arg\,min}_{k \in \{1,\ldots,K\}} \|x_n - \mu_k\|^2.}$$

Because the square function is monotonic for nonnegative values, minimizing squared distance gives the same nearest-centroid assignment as minimizing Euclidean distance. The assignment boundaries between two centres are perpendicular bisectors, producing Voronoi regions.

2.4 Deriving the centroid update

Now hold assignments fixed. The objective for cluster k is

$$J_k(\mu_k) = \sum_{n : c_n = k} \|x_n - \mu_k\|^2.$$

Expand the squared norm:

$$J_k(\mu_k) = \sum_{n : c_n = k} (x_n^T x_n - 2 x_n^T \mu_k + \mu_k^T \mu_k).$$

Differentiate with respect to μ_k:

$$\nabla_{\mu_k} J_k = -2 \sum_{n : c_n = k} x_n + 2 N_k \mu_k,$$

where $N_k = \#\{n : c_n = k\}$. Set the gradient to zero:

$$-2 \sum_{n : c_n = k} x_n + 2 N_k \mu_k = 0.$$

Therefore,

$$\boxed{\mu_k \leftarrow \frac{1}{N_k} \sum_{n : c_n = k} x_n.}$$

The Hessian is $2N_kI$, positive definite when $N_k > 0$, so this is the minimizing mean for a nonempty cluster.

Result to remember

Why the mean? Under squared Euclidean loss, the arithmetic mean is the point minimizing the sum of squared distances. Under absolute-distance loss in one dimension, the median plays the analogous role.

2.5 The K-means algorithm

1. Choose K initial centroids (randomly or with K-means++).
2. **Assignment:** assign each point to its nearest centroid.
3. **Update:** replace each centroid by the mean of its assigned points.
4. Recompute the objective J and repeat steps 2–3 until convergence.

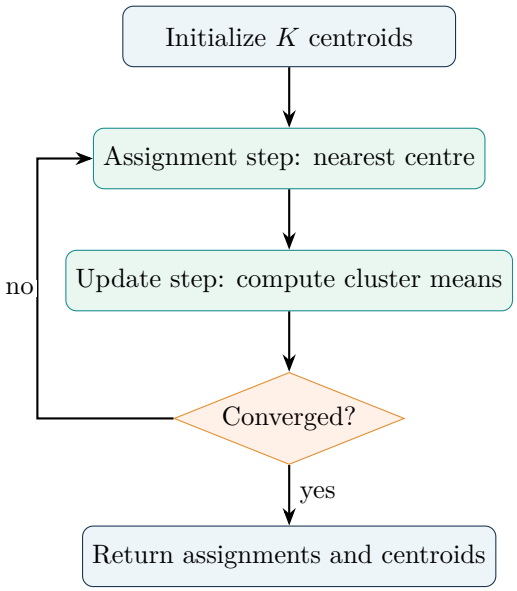


Figure 2: K-means alternates two conditional minimization steps.

2.6 Why the objective never increases

During assignment, each point is reassigned to a centre no farther away than its previous centre. Therefore J cannot increase. During the update step, each centroid is replaced by the mean, which minimizes the fixed-assignment squared-error objective. Thus

$$J^{(t+1)} \leq J^{(t)}.$$

Since $J \geq 0$, the objective values converge. With a finite set of possible assignments and a deterministic tie-breaking rule, the algorithm eventually reaches a fixed assignment (ignoring degenerate implementation details).

Common trap

Monotonic decrease does *not* imply that K-means finds the global minimum. The objective is jointly non-convex in assignments and centres. Initialization can lead to different local minima.

2.7 Worked numerical example

Consider one-dimensional data $X = \{1, 2, 8, 9\}$ with $K = 2$ and initial centres $\mu_1 = 1$, $\mu_2 = 8$.

1. Assignment: $1, 2$ go to cluster 1; $8, 9$ go to cluster 2.
2. Update:
$$\mu_1 = \frac{1+2}{2} = 1.5, \qquad \mu_2 = \frac{8+9}{2} = 8.5.$$
3. Distortion:
$$J = (1-1.5)^2 + (2-1.5)^2 + (8-8.5)^2 + (9-8.5)^2 = 1.$$

The assignments remain unchanged, so this solution is stable.

2.8 Initialization, empty clusters, and choosing K

Random initialization can place centres poorly or close together. **K-means++** chooses a first centre randomly and then samples later centres with probability proportional to squared distance from the nearest existing centre; this tends to spread initial centres apart.

An **empty cluster** occurs if no points are assigned to a centre. Common remedies are to reinitialize it at a poorly represented point, split a large cluster, or use an implementation-specific recovery rule.

The objective always decreases as K grows (or at least does not increase), so training distortion alone cannot choose K. The **elbow method** plots distortion against K and looks for a point after which improvement slows. Silhouette score, stability, domain knowledge, and downstream usefulness can provide additional evidence.

2.9 Strengths and limitations

Strength	Limitation
Simple and computationally efficient	Must choose K in advance
Easy to explain and implement	Sensitive to initialization and outliers
Scales well to many observations	Hard assignments ignore uncertainty
Works well for compact, separated groups	Favours roughly spherical/equal-scale clusters under Euclidean distance
Centres provide a compact representation	Mean can be misleading for non-convex or highly unequal-density groups

3 Why move from K-means to a GMM?

K-means makes several strong modelling choices: one representative centre per cluster, Euclidean distance, and a hard assignment. These assumptions can be too restrictive.

3.1 Hard membership versus uncertainty

K-means says a point belongs to exactly one cluster. But a point near the boundary between two populations may plausibly belong to either. A probabilistic model can return, for example, $P(z_n = 1 \mid x_n) = 0.52$ and $P(z_n = 2 \mid x_n) = 0.48$, rather than forcing certainty.

3.2 Spherical distance versus covariance-aware shape

K-means uses distance to the centre and does not explicitly learn cluster covariance. A GMM can represent elongated, tilted, or differently sized clusters through covariance matrices. The covariance controls the geometry of equal-density contours.

3.3 From prototype to distribution

K-means represents component k by μ_k. A GMM represents it using a weight π_k, mean μ_k, and covariance Σ_k. Instead of asking only "which centre is nearest?", it asks "how likely is this observation under each component, accounting for that component's spread and orientation?"

Key idea

A GMM is not automatically superior. It is more expressive, but it estimates more parameters, may overfit with limited data, and can suffer from singular covariance estimates. Use the richer model when probabilistic membership or non-spherical Gaussian structure is useful.

4 Gaussian Mixture Models (GMMs)

4.1 The mixture model

Assume each observation is generated by a two-stage process:

1. Choose a latent component $z_n = k$ with probability π_k.
2. Draw x_n from a Gaussian distribution specific to component k.

The mixing weights satisfy

$$\pi_k \geq 0, \qquad \sum_{k=1}^{K} \pi_k = 1.$$

For component k,

$$x_n \mid z_n = k \sim \mathcal{N}(\mu_k, \Sigma_k).$$

Marginalizing over the unknown component gives

$$\boxed{p(x_n) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x_n \mid \mu_k, \Sigma_k).}$$

For d dimensions,

$$\mathcal{N}(x \mid \mu, \Sigma) = \frac{\exp\left[-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right]}{(2\pi)^{d/2}|\Sigma|^{1/2}}.$$

4.2 What each parameter means

- π_k: prior proportion or mixture weight of component k.

- $\mu_k \in \mathbb{R}^d$: centre of component k.
- $\Sigma_k \in \mathbb{R}^{d\times d}$: covariance, controlling scale, shape, and orientation.

A full covariance has $d(d+1)/2$ unique entries. Diagonal covariance assumes uncorrelated dimensions within a component; spherical covariance uses $\Sigma_k = \sigma_k^2 I$.

4.3 The joint distribution and latent indicators

Let z_n be a one-hot vector, where $z_{nk} = 1$ means point n came from component k. Then

$$p(z_n) = \prod_{k=1}^{K} \pi_k^{z_{nk}}, \qquad p(x_n \mid z_n) = \prod_{k=1}^{K} \mathcal{N}(x_n \mid \mu_k, \Sigma_k)^{z_{nk}}.$$

Therefore,

$$\boxed{p(x_n, z_n) = \prod_{k=1}^{K} \left[\pi_k \mathcal{N}(x_n \mid \mu_k, \Sigma_k)\right]^{z_{nk}}.}$$

For independent observations, $p(X, Z) = \prod_n p(x_n, z_n)$.

4.4 Likelihood and log-likelihood

The observed-data likelihood is

$$p(X \mid \theta) = \prod_{n=1}^{N} \sum_{k=1}^{K} \pi_k \mathcal{N}(x_n \mid \mu_k, \Sigma_k),$$

where $\theta = \{\pi_k, \mu_k, \Sigma_k\}_{k=1}^K$. Hence

$$\boxed{\ell(\theta) = \sum_{n=1}^{N} \log \left[\sum_{k=1}^{K} \pi_k \mathcal{N}(x_n \mid \mu_k, \Sigma_k)\right].}$$

The logarithm is outside the sum. This coupling is crucial: the log-likelihood does not separate into independent simple maximizations for each component.

Common trap

In general, $\log \sum_k a_k \neq \sum_k \log a_k$. Do not move the logarithm inside the mixture sum. This is one central reason the direct MLE is difficult.

4.5 Posterior responsibilities

Using Bayes' theorem,

$$r_{nk} := P(z_n = k \mid x_n, \theta) = \frac{p(x_n \mid z_n = k, \theta) P(z_n = k \mid \theta)}{\sum_j p(x_n \mid z_n = j, \theta) P(z_n = j \mid \theta)}.$$

Thus,

$$\boxed{r_{nk} = \frac{\pi_k \mathcal{N}(x_n \mid \mu_k, \Sigma_k)}{\sum_{j=1}^{K} \pi_j \mathcal{N}(x_n \mid \mu_j, \Sigma_j)}.}$$

For each point, $0 \le r_{nk} \le 1$ and $\sum_k r_{nk} = 1$. These are soft assignments: every component contributes a fractional responsibility.

4.6 Illustrative density picture

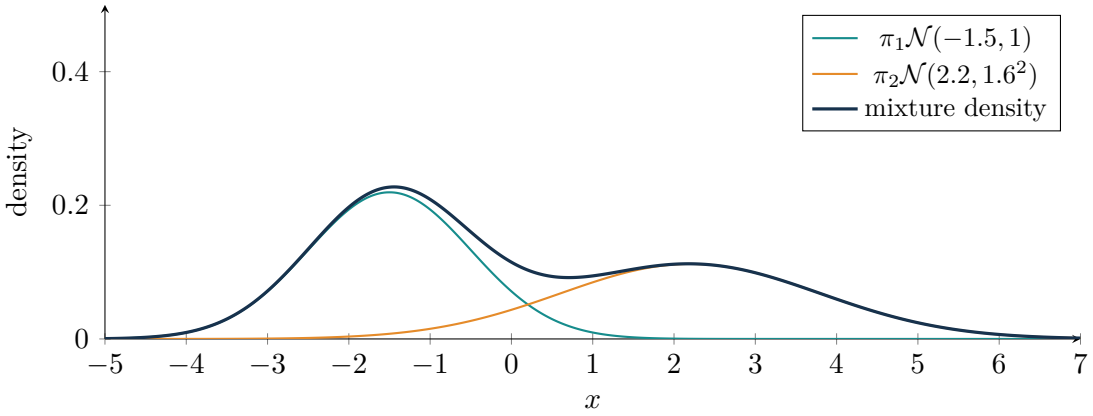


Figure 3: A mixture density is the weighted sum of component densities; it need not itself be a single Gaussian.

5 Expectation-Maximization (EM)

5.1 Why EM is needed

The mixture likelihood contains a sum over hidden component identities inside a logarithm. If the component assignment of each point were known, parameter estimation would be much easier: means and covariances could be estimated from the assigned observations. But assignments are unknown. EM alternates between estimating soft assignments using current parameters and re-estimating parameters using those soft assignments.

Key idea

E-step: estimate how responsible each component is for each observation, given current parameters.
M-step: maximize the expected complete-data log-likelihood, treating those responsibilities as fixed weights.

5.2 Complete data versus observed data

Observed data are $X = \{x_n\}_{n=1}^N$. The latent variables $Z = \{z_n\}_{n=1}^N$ encode the unobserved component memberships. If both X and Z were observed, the complete-data log-likelihood would be

$$\log p(X, Z \mid \theta) = \sum_{n=1}^{N} \sum_{k=1}^{K} z_{nk} \left[\log \pi_k + \log \mathcal{N}(x_n \mid \mu_k, \Sigma_k)\right].$$

But we do not know z_{nk}. EM replaces each hidden indicator by its current conditional expectation:

$$\mathbb{E}[z_{nk} \mid x_n, \theta^{\text{old}}] = P(z_n = k \mid x_n, \theta^{\text{old}}) = r_{nk}.$$

5.3 E-step: compute responsibilities

At iteration t, using current parameters $\theta^{(t)}$:

$$\boxed{r_{nk}^{(t)} = \frac{\pi_k^{(t)} \mathcal{N}(x_n \mid \mu_k^{(t)}, \Sigma_k^{(t)})}{\sum_j \pi_j^{(t)} \mathcal{N}(x_n \mid \mu_j^{(t)}, \Sigma_j^{(t)})}.}$$

Intuition: a component receives high responsibility when its prior weight is large and the observation lies in a high-density region under that component.

5.4 M-step: maximize the expected complete-data log-likelihood

Define the effective membership count

$$N_k = \sum_{n=1}^{N} r_{nk}.$$

The expected complete-data log-likelihood, often called Q, is

$$Q(\theta, \theta^{(t)}) = \sum_n \sum_k r_{nk}^{(t)} \left[\log \pi_k + \log \mathcal{N}(x_n \mid \mu_k, \Sigma_k)\right].$$

Maximizing this expression yields the parameter updates below.

5.4.1 Mixing-weight update

The terms involving π_k are $\sum_k N_k \log \pi_k$, subject to $\sum_k \pi_k = 1$. Use a Lagrange multiplier:

$$\mathcal{L} = \sum_k N_k \log \pi_k + \lambda \left(\sum_k \pi_k - 1 \right).$$

Then

$$\frac{\partial \mathcal{L}}{\partial \pi_k} = \frac{N_k}{\pi_k} + \lambda = 0 \quad \Longrightarrow \quad \pi_k = -\frac{N_k}{\lambda}.$$

Summing over k gives $1 = -\frac{\sum_k N_k}{\lambda} = -\frac{N}{\lambda}$, so $\lambda = -N$ and

$$\boxed{\pi_k^{\text{new}} = \frac{N_k}{N}.}$$

5.4.2 Mean update

The Gaussian log density contributes a quadratic term in μ_k. Up to terms independent of μ_k,

$$Q_{\mu_k} = -\frac{1}{2} \sum_n r_{nk} (x_n - \mu_k)^T \Sigma_k^{-1} (x_n - \mu_k).$$

Differentiate:

$$\nabla_{\mu_k} Q_{\mu_k} = \sum_n r_{nk} \Sigma_k^{-1} (x_n - \mu_k) = 0.$$

Multiplying by Σ_k and rearranging,

$$\sum_n r_{nk} x_n - \mu_k \sum_n r_{nk} = 0,$$

hence

$$\boxed{\mu_k^{\text{new}} = \frac{\sum_n r_{nk} x_n}{N_k}.}$$

This is a weighted mean. Points with high responsibility influence the mean more strongly.

5.4.3 Covariance update

For a full covariance, the relevant terms are

$$Q_{\Sigma_k} = -\frac{1}{2}\sum_n r_{nk}\left[\log|\Sigma_k| + (x_n-\mu_k)^T\Sigma_k^{-1}(x_n-\mu_k)\right] + C.$$

The maximizer is the weighted covariance around the newly updated mean:

$$\boxed{\Sigma_k^{\text{new}} = \frac{1}{N_k}\sum_n r_{nk}(x_n-\mu_k^{\text{new}})(x_n-\mu_k^{\text{new}})^T.}$$

The denominator is N_k, the effective component count. In practical implementations, a small diagonal regularizer λI may be added to avoid singular covariance matrices.

Result to remember

GMM EM updates at a glance

$$r_{nk} = \frac{\pi_k\mathcal{N}(x_n \mid \mu_k,\Sigma_k)}{\sum_j \pi_j\mathcal{N}(x_n \mid \mu_j,\Sigma_j)}, \quad N_k = \sum_n r_{nk}$$

$$\pi_k \leftarrow N_k/N, \qquad \mu_k \leftarrow \frac{\sum_n r_{nk}x_n}{N_k}$$

$$\Sigma_k \leftarrow \frac{\sum_n r_{nk}(x_n-\mu_k)(x_n-\mu_k)^T}{N_k}.$$

5.5 The complete EM loop

1. Initialize π_k, μ_k, Σ_k.
2. E-step: compute all responsibilities r_{nk}.
3. M-step: compute N_k and update π_k, μ_k, Σ_k.
4. Evaluate the observed-data log-likelihood $\ell(\theta)$.
5. Repeat until the log-likelihood improvement is small or a maximum iteration count is reached.

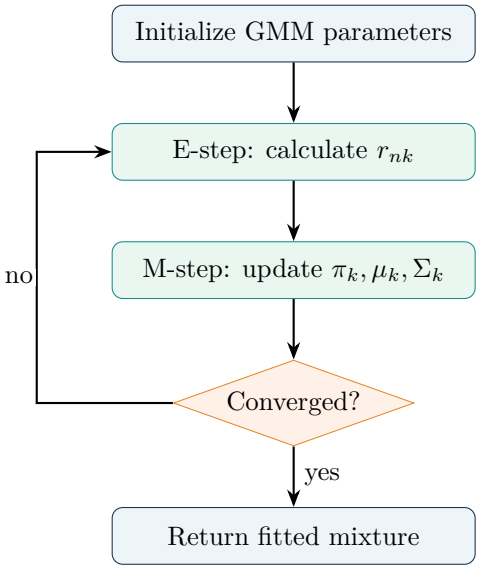


Figure 4: EM alternates inference over latent memberships and parameter optimization.

5.6 A tiny responsibility example

Suppose one observation has component weighted densities

$$\pi_1 \mathcal{N}(x \mid \mu_1, \Sigma_1) = 0.30, \qquad \pi_2 \mathcal{N}(x \mid \mu_2, \Sigma_2) = 0.20.$$

The marginal density is $p(x) = 0.30 + 0.20 = 0.50$, so

$$r_1 = 0.30/0.50 = 0.60, \qquad r_2 = 0.20/0.50 = 0.40.$$

The observation contributes 0.60 effective membership to component 1 and 0.40 to component 2, rather than being assigned wholly to either component.

6 Why EM increases likelihood: the lower-bound view

This section explains the general EM idea, beyond the GMM formulas.

6.1 Marginal likelihood with latent variables

Let X be observed data, Z latent variables, and θ model parameters. The observed-data likelihood is

$$p(X \mid \theta) = \sum_Z p(X, Z \mid \theta),$$

where the sum becomes an integral for continuous latent variables. Direct maximization may be hard because latent states have been summed out.

6.2 Introduce an auxiliary distribution

Let $q(Z)$ be any distribution over latent states. Since

$$\log p(X \mid \theta) = \log \sum_Z q(Z) \frac{p(X, Z \mid \theta)}{q(Z)},$$

Jensen's inequality gives

$$\log p(X \mid \theta) \geq \sum_Z q(Z) \log \frac{p(X, Z \mid \theta)}{q(Z)} =: \mathcal{L}(q, \theta).$$

The quantity $\mathcal{L}(q, \theta)$ is a lower bound on the log-likelihood, also called the ELBO in variational-inference contexts.

6.3 Exact identity and the KL divergence

Using the definition

$$\mathrm{KL}(q \| p) = \sum_Z q(Z) \log \frac{q(Z)}{p(Z)},$$

one can derive the identity

$$\boxed{\log p(X \mid \theta) = \mathcal{L}(q, \theta) + \mathrm{KL}(q(Z) \,\|\, p(Z \mid X, \theta))\,.}$$

The KL divergence is nonnegative. Therefore the lower bound is tight when

$$q(Z) = p(Z \mid X, \theta).$$

6.4 E-step and M-step from the bound

E-step: with $\theta = \theta^{(t)}$, choose

$$q^{(t)}(Z) = p(Z \mid X, \theta^{(t)}),$$

which makes the bound tight at the current parameter value.

M-step: hold $q^{(t)}$ fixed and choose

$$\theta^{(t+1)} = \arg\max_{\theta} \mathcal{L}(q^{(t)}, \theta).$$

Because the entropy term $-\sum_Z q^{(t)}(Z) \log q^{(t)}(Z)$ is constant with respect to θ, this is equivalent to

$$\boxed{\theta^{(t+1)} = \arg\max_{\theta} \mathbb{E}_{Z|X,\theta^{(t)}}[\log p(X, Z \mid \theta)].}$$

This is the general EM update.

Common trap

EM guarantees non-decrease of the observed-data log-likelihood under exact E- and M-steps. It does not guarantee a global maximum, a unique solution, or that every parameter sequence is well-behaved in degenerate models.

7 The relationship between K-means and GMMs

K-means can be understood as a limiting special case of a Gaussian mixture under restrictive assumptions. Suppose all components have equal mixing weights and a shared spherical covariance:

$$\pi_k = 1/K, \qquad \Sigma_k = \sigma^2 I.$$

Then the responsibility is proportional to

$$r_{nk} \propto \exp\left(-\frac{\|x_n - \mu_k\|^2}{2\sigma^2}\right).$$

As $\sigma^2 \to 0$, the nearest mean dominates the exponential. Except for exact ties, the responsibility of the closest component approaches 1 and all others approach 0. Thus the soft assignments become hard assignments.

Under the same limit, maximizing the relevant expected complete-data log-likelihood corresponds to minimizing

$$\sum_n \|x_n - \mu_{c_n}\|^2,$$

the K-means distortion, up to constants and positive scaling. This explains why K-means and GMMs are closely related, while also clarifying that a general GMM is more flexible.

Property	K-means	GMM + EM
Membership	Hard, one cluster per point	Soft posterior responsibilities
Cluster representation	Centroid	Weight, mean, covariance
Geometry	Euclidean / Voronoi regions	Covariance-aware Gaussian ellipsoids
Objective	Minimize squared distortion	Increase observed-data log-likelihood
Uncertainty	Not directly represented	Explicit posterior membership probabilities
Typical cost	Often lower	Higher, especially with full covariances

8 Practical issues and model selection

8.1 Initialization and local optima

EM, like K-means, can converge to a local optimum or stationary point. A common strategy is to run multiple initializations and keep the solution with the highest final log-likelihood. K-means centres are often used to initialize GMM means; covariances can be initialized from assigned points with regularization.

8.2 Covariance collapse and numerical stability

A Gaussian component can sometimes place its mean on a data point while its covariance shrinks toward zero, making the likelihood arbitrarily large in unconstrained full-covariance mixtures. Practical implementations may impose covariance floors, add λI, constrain covariance structure, or use Bayesian priors.

For numerical stability, compute mixture log probabilities with the log-sum-exp trick:

$$\log \sum_k e^{a_k} = m + \log \sum_k e^{a_k - m}, \qquad m = \max_k a_k.$$

This avoids underflow when Gaussian densities are extremely small.

8.3 Choosing the number of components

The number K can be compared using held-out likelihood, stability, domain knowledge, or information criteria. For a fitted model with p free parameters and maximized log-likelihood $\hat{\ell}$,

$$\mathrm{AIC} = 2p - 2\hat{\ell}, \qquad \mathrm{BIC} = p \log N - 2\hat{\ell}.$$

Lower AIC/BIC is preferred among compared models under the respective criteria. For a full-covariance GMM in d dimensions, the parameter count is

$$p = (K-1) + Kd + K\frac{d(d+1)}{2}.$$

The terms correspond to independent mixing weights, means, and symmetric covariance matrices. This count assumes no additional constraints or tied parameters.

9 Applications: when should each method be used?

- **Use K-means** when you need a fast baseline, compact prototypes, and clusters are reasonably compact under the chosen feature scaling and distance.
- **Use a GMM** when soft membership is meaningful, clusters have different orientations or scales, or density estimates are useful.
- **Use neither blindly** when groups are non-Gaussian, non-convex, strongly density-connected, or when outliers dominate. Consider density-based or hierarchical methods, depending on the problem.

10 Common misconceptions and exam traps

1. **"K-means always finds the best clustering."** False. It converges to a local solution dependent on initialization.
2. **"The centroid must be a data point."** False. It is the arithmetic mean and can lie between observations.
3. **"A lower training distortion proves a better model."** False. Distortion tends to fall as K increases; complexity and usefulness matter.
4. **"GMM responsibilities are labels."** They are posterior probabilities; a hard label can be produced later using $\arg\max_k r_{nk}$ if needed.
5. **"EM assigns a point once and never revisits it."** False. Responsibilities are recomputed each E-step as parameters change.
6. **"EM always finds the global maximum."** False. Likelihood is non-decreasing under standard exact EM, but local optima and degeneracies remain.
7. **"The GMM likelihood is a sum of log component likelihoods."** False. It is a sum of logs of mixture sums: $\sum_n \log \sum_k \pi_k \mathcal{N}(\cdot)$.
8. **"K-means and GMM are unrelated."** False. K-means appears as a hard-assignment limit under equal-weight, shared spherical Gaussians with variance approaching zero.
9. **"More components always means a more meaningful model."** False. More components improve flexibility but can overfit or represent noise.

11 One-page formula sheet

K-means objective:

$$J = \sum_n \|x_n - \mu_{c_n}\|^2 = \sum_n \sum_k r_{nk} \|x_n - \mu_k\|^2.$$

Assignment and update:

$$c_n \leftarrow \arg\min_k \|x_n - \mu_k\|^2, \qquad \mu_k \leftarrow \frac{1}{N_k} \sum_{n:c_n=k} x_n.$$

GMM density and responsibilities:

$$p(x_n) = \sum_k \pi_k \mathcal{N}(x_n \mid \mu_k, \Sigma_k), \qquad r_{nk} = \frac{\pi_k \mathcal{N}(x_n \mid \mu_k, \Sigma_k)}{\sum_j \pi_j \mathcal{N}(x_n \mid \mu_j, \Sigma_j)}.$$

EM effective counts and updates:

$$N_k = \sum_n r_{nk}, \quad \pi_k = \frac{N_k}{N}, \quad \mu_k = \frac{\sum_n r_{nk} x_n}{N_k},$$

$$\Sigma_k = \frac{\sum_n r_{nk}(x_n - \mu_k)(x_n - \mu_k)^T}{N_k}.$$

General EM:

$$Q(\theta, \theta^{old}) = \mathbb{E}_{Z|X,\theta^{old}}[\log p(X, Z \mid \theta)], \quad \theta^{new} = \arg\max_{\theta} Q(\theta, \theta^{old}).$$

Lower-bound identity:

$$\log p(X \mid \theta) = \mathcal{L}(q, \theta) + \mathrm{KL}(q(Z) \| p(Z \mid X, \theta)).$$

Model selection:

$$\mathrm{AIC} = 2p - 2\hat{\ell}, \qquad \mathrm{BIC} = p \log N - 2\hat{\ell}.$$

12 Final study strategy

1. Derive the K-means centroid update by differentiating the squared-error objective.
2. Explain why each K-means step does not increase distortion, and why this is not a global-optimum guarantee.
3. State the modelling assumptions that make K-means restrictive.
4. Derive the GMM marginal density by summing over the latent component identity.
5. Derive responsibilities directly from Bayes' theorem.
6. Memorize and understand the three GMM M-step updates: mixture weights, means, covariances.
7. Explain the E-step/M-step interpretation using observed and latent variables.
8. Reproduce the Jensen/KL lower-bound argument and explain likelihood monotonicity.
9. Explain the hard-assignment limit connecting GMMs to K-means.
10. Know initialization, covariance collapse, numerical stability, and model-selection caveats.

Further reading

- Chloe Bee, "The EM Algorithm Explained," Medium: read the article here.
- Bishop, C. M., *Pattern Recognition and Machine Learning*, chapters on mixture models and EM.
- Dempster, A. P., Laird, N. M., and Rubin, D. B. (1977), "Maximum Likelihood from Incomplete Data via the EM Algorithm," *Journal of the Royal Statistical Society: Series B*.

Understand the assumptions; derive the updates; interpret the result.